

Chapter 17

Agriculture — IV: Cash Crops (2)

Class 10 - Total Geography Morning Star

Choose the correct option

Question 1

Cotton is tropical crop that requires uniformly high

1. rainfall
2. temperature
3. humidity
4. All of the above.

Answer

temperature

Question 2

Cotton plant is extremely sensitive to which can kill the plant and destroy the crop.

1. temperature
2. rainfall
3. frost
4. humidity

Answer

frost

Question 3

..... soil is ideal for cotton cultivation.

1. Alluvium
2. Black

3. Laterite
4. Red

Answer

Black

Question 4

The fibre or lint is separated from the cotton seeds by a process known as

1. ginning
2. spinning
3. pruning
4. deseeding

Answer

ginning

Question 5

An inferior substitute for jute which is more coarse and less strong than jute is

1. Testa
2. Mesta
3. Cesta
4. Jesta

Answer

Mesta

Question 6

The soil gives the heaviest yield of Jute.

1. clayey
2. loamy
3. siety
4. sandy

Answer

clayey

Question 7

..... softens the outer bark of jute and facilitates the early removability of the fibre within.

1. Jetting
2. Retting
3. Metting
4. Netting

Answer

Retting

Question 8

Tea develops most vigorously under and no

1. sunshine, rains
2. light shade, dry spells
3. moisture, high temperature
4. clear sky, no rains

Answer

light shade, dry spells

Question 9

High humidity, heavy and morning are good for the rapid development of young tea leaves.

1. dew, fog
2. rainfall, sunshine
3. rainfall, dew
4. shade, fog

Answer

dew, fog

Question 10

The process of growing tea shrubs in nurseries from cutting of high yielding varieties is known as

1. clonal sapling
2. clonal planting
3. clonal cuttings
4. clonal trimmings

Answer

clonal planting

Question 11

The tea shrub continues to be productive for about years.

1. 20
2. 25
3. 50
4. 70

Answer

50

Question 12

..... variety of tea is greenish brown in colour.

1. Oolong
2. Brick
3. Black
4. Green

Answer

Oolong

Question 13

..... variety of tea is normally consumed in Russia and Tibet.

1. Black
2. Brick

3. Oolong
4. Green

Answer

Brick

Question 14

..... has the largest area under tea production in the world.

1. China
2. Sri Lanka
3. India
4. Kenya

Answer

India

Question 15

..... is the largest producer of tea in India.

1. Tamil Nadu
2. Kerala
3. West Bengal
4. Assam

Answer

Assam

Question 16

Which of the following are non-traditional coffee producing areas?

1. Andhra Pradesh and Odisha
2. Kerala and Karnataka
3. Manipur and Tripura
4. Bihar and Punjab

Answer

Answer the following questions

Question 1(a)

What are fibre crops? Give two examples of plant fibre crops.

Answer

Fibre crops are crops that are primarily grown for obtaining fibre. The fibre is traditionally used to make paper, cloth, or rope.

Two fibre crops are cotton and jute.

Question 1(b)

What conditions of soil favour the growth of cotton? Why?

Answer

Cotton grows well in the well-drained clayey soils containing lime and phosphates. The deep and medium black soils are considered ideal for cotton cultivation since black soil has a high level of clay and good water holding capacity that makes it suitable for the growth of cotton.

Since cotton crops exhaust the fertility of soil rapidly, regular application of manures and fertilizers to the soil is necessary.

Question 1(c)

(i) Name the two chief cotton-growing areas in India.

(ii) Which climatic conditions favour the cultivation of cotton?

Answer

(i) Two chief cotton-growing areas in India are-

1. The north western Deccan on the fertile Black cotton soil
2. The central and southern Deccan of Karnataka and Tamil Nadu

(ii) The climatic conditions that favour the cultivation of cotton are:

1. **Temperature** — Cotton requires uniformly high temperatures between 21°C and 30°C. A long growing period of at least 200 frost-free days is required for the plant to mature.

2. **Rainfall** — Rainfall should be moderate, ranging between 50 to 75 cm. Sufficient rainfall is required during the early stages of the growth, but a sunny and dry weather is required after flowering.

Question 1(d)

- (i) How is frost harmful for the growth of cotton plant?
- (ii) Why is dry weather necessary at the time of harvesting cotton?

Answer

- (i) The cotton plant is extremely sensitive to frost and late spring or early autumn frosts can kill the plant and destroy the crop. The soil temperature below 20°C reduces root growth thus retarding plant growth.
- (ii) Dry weather is necessary at the time of harvesting cotton as rainfall during boll-opening and harvesting periods is harmful for the plants as it makes them vulnerable to pests and diseases.

Question 2(a)

What advantages does Long Staple Cotton have?

Answer

The advantages of Long Staple Cotton is that the longer the cotton fiber, the stronger, softer, and more durable the resulting fabric. Long staple cotton is long, fine and shiny. It is used for making fine and superior quality cloth.

Fabrics made of long-staple cottons fray less, pill less, wrinkle less, and even fade less than fabrics made with their short-staple counterparts.

Question 2(b)

Why Gujarat and Maharashtra are the leading producers of Cotton?

Answer

Gujarat and Maharashtra are the leading producers of Cotton because the moist climate of both the states is well suited for cotton cultivation and the temperature is ideal for the cotton industries as the cotton threads tend to break in dry climate while they rarely break in moist and humid climate.

Question 2(c)

Describe the process or stages in Ginning of Cotton.

Answer

The stages in Ginning of Cotton are-

1. The cotton first goes through the dryers to reduce moisture content.
2. Then, it goes through the cleaning equipment to remove foreign matter.
3. The cotton is then air conveyed to gin stands where its circular saws pull the lint through closely spaced ribs that prevent the seed from passing through.
4. The lint is removed from the saw teeth by rotating brushes and then compressed into bales.

Question 2(d)

What are the problems associated with the growing of cotton?

Answer

The problems associated with the growing of cotton are-

1. Lack of financial facility
2. Shortage of labour
3. Inadequate water supply
4. Severity of diseases
5. Lack of technology
6. High cost of inputs
7. Low quality of fertilizers and pesticides
8. Poor quality of seeds

Question 3(a)

State the soil conditions that favour the growth of jute.

Answer

Jute grows best on the soil enriched by new alluvium brought by river inundation. Loamy soils are most suitable for jute production as the clay particles of loam help to hold the plant nutrients preventing them from getting washed away by water.

The clayey soil gives the heaviest yield. Sandy soils produce coarse fibres.

Question 3(b)

In what way is Ganga-Brahmaputra Delta suitable for jute cultivation?

Answer

Ganga-Brahmaputra Delta has loamy soil which is most suitable for jute production as the clay particles of loam help to hold the plant nutrients preventing them from getting washed away by water.

Question 3(c)

(i) Why is jute retted?

(ii) How is this done?

Answer

(i) Jute is retted because retting softens the outer bark and facilitates the early removal of the fibre within.

(ii) Retting was once done by submerging the jute into ponds and streams but is now done in special tanks. Chemical additives help in the retting operation.

Question 3(d)

Describe briefly the processing of jute.

Answer

Jute is harvested by hand, by pulling up the stem. It is dried and stripped of unwanted leaves and is put in water and allowed to rot. This process is known as retting.

Retting was once done by submerging the jute into ponds and streams but is now done in special tanks. Chemical additives help in the retting operation.

The fleshy part of the stem is eventually decomposed and the fibres are then scraped to remove any remaining pieces of soft vegetable matter. After drying, the fibres are loosely spun and woven, and are used in making sacks and bags, carpet, upholstery, etc.

Question 4(a)

Why is mesta an inferior substitute for jute? Where is it grown?

Answer

Mesta is an inferior substitute for jute as the fibre is more coarse and not as strong as jute fibre.

Mesta is cultivated in some parts of Assam, Bihar, Odisha and Kerala.

Question 4(b)

By what other name is jute referred to? Why?

Answer

Jute is also called 'Golden fibre' as it provides huge revenue to the government.

Question 4(c)

- (i) Why are floods beneficial for the growth of jute?
- (ii) Mention one advantage and one disadvantage of delay in jute harvest.

Answer

- (i) Floods are beneficial for the growth of jute because plenty of water is needed for the process of retting and a new soil cover is available.
- (ii) One advantage of delay in jute harvest is that it adds to the yield and one disadvantage of delay in jute harvest is that it usually produces coarse fibre.

Question 4(d)

State any three uses of jute.

Answer

Three uses of jute are-

1. It is used for manufacturing rough quality cloth, sacks and other packaging material.
2. It is used for making many utility products like carpets, rugs, etc.
3. It provides a huge revenue to the government.

Question 5(a)

State any two conditions that favour the growth of tea in the Nilgiris.

Answer

Two conditions that favour the growth of tea in the Nilgiris are-

1. The slopes protect the crop from annual inundations and stagnant water during the rains.
2. The high altitude provides the required climatic conditions- temperature ranging from 24°C to 30°C and annual rainfall of atleast 150 cm, well distributed throughout the year.

Question 5(b)

State the advantages of growing tea plants on hill slopes.

Answer

Tea plantations require even distribution of water without any water logging. Hill slopes provide proper drainage and prevents water logging problems. This is the reason why tea plantations are usually grown on hill slopes.

Question 5(c)

(i) Name the different varieties of tea grown in India.

(ii) Why is blending necessary for tea?

Answer

(i) The different varieties of tea grown in India are-

1. Black tea
2. Green tea
3. Oolong tea
4. Brick tea

(ii) Blending is necessary for tea to give it a special aroma and make many proprietary brands.

Question 5(d)

(i) Name the leading producer of tea in India. State two factors that have helped it to become the leading state in tea production.

(ii) Define the following terms:

1. Clonal Planting
2. Pruning

Answer

(i) Assam is the leading producer of tea in India.

Two factors that have helped Assam to become the leading state in tea production are-

1. No stagnant water due to hilly terrain
2. Availability of cheap labor

(ii) The definitions of Clonal Planting & Pruning are given below:

1. **Clonal Planting** — Tea shrubs can be grown in nurseries from cuttings of high yielding varieties. This is known as clonal planting.

2. **Pruning** — Pruning means plucking of tea leaves. Frequent pruning encourages the rapid production of fresh leaves and shoots. Pruning of tea bush starts after two years in order to maintain the height and diameter of the plant limited to one metre.

Question 6(a)

Give two climatic factors that favour the cultivation of coffee.

Answer

Two climatic factors that favour the cultivation of coffee are-

1. **Temperature** — Coffee plant requires warm climate with temperature ranging from 15°C to 28°C and a moderate supply of moisture.
2. **Rainfall** — During the period of growth, the plant requires 150 - 200 cm of annual rainfall.

Question 6(b)

Name the three varieties of coffee plants grown on commercial scale in India. Name one state where coffee is grown extensively.

Answer

The three varieties of coffee plants grown on commercial scale in India are-

1. Coffee Robusta
2. Coffee Liberica
3. Coffee Arabica

Coffee is grown extensively in Karnataka.

Question 6(c)

Give a geographical reason for each of the following:

- (i) Coffee is grown on the slopes of the hills in Peninsular India.
- (ii) Coffee estates have coffee inter-planted with orange trees, cardamom and pepper vines.
- (iii) Tea bushes are pruned at regular intervals.

Answer

(i) Coffee is grown on the slopes of the hills in Peninsular India to avoid water logging and to supply even water to the plants. Soils are also fertile in this area with sufficient amount of rainfall of about 150 cm on average, necessary for the growth of coffee plantations.

(ii) Coffee plant is susceptible to direct sunrays. So, Coffee estates have coffee inter-planted with orange trees, cardamom and pepper vines to provide shade to the coffee plant and at the same time generate extra income for the farmers.

(iii) Tea bushes are pruned at regular interval to maintain the height and diameter of the plant limited to one metre. Frequent pruning encourages the rapid production of fresh leaves and shoots.

Question 6(d)

Describe briefly the Dry Parchment method of processing of coffee.

Answer

In the Dry Parchment method of processing of coffee, the following process is followed-

1. The harvested cherries are sorted and cleaned to separate the unripe, overripe and damaged cherries and remove dirt, soil, twigs and leaves.
2. The coffee cherries are then spread out in the sun to dry.
3. The beans are then fermented by drying in the sun for a week.
4. After drying, the machines peel off the two layers of inner husks.
5. They are sorted according to size and quality and then packed in sacks for use.
6. The beans are roasted at temperatures of about 99°C and then grounded into coffee powder which is used to make beverage. Roasting gives it brown colour and characteristic aroma and taste.

Thinking Skills

Question 1

Cotton and Jute are the two crops which can provide employment to a large number of people and enhance the foreign exchange of the country. Give your opinion on the statement.

Answer

I agree that cotton and jute are two crops that can provide employment to a large number of people and enhance the foreign exchange of India.

Both cotton and jute cultivation and related industries have the potential to generate employment opportunities, particularly in rural areas. These crops require labor-intensive activities such as

sowing, harvesting, and processing, which can provide jobs to a significant number of people, especially during peak seasons.

Cotton and jute can contribute to foreign exchange earnings through their exports. Cotton is a valuable raw material for textile industries, and jute is used for manufacturing various products like sacks, bags, and carpets. The export of cotton textiles and jute goods can generate foreign currency for the country.

With proper attention to improving productivity, enhancing value addition, promoting innovation, and addressing challenges, the cotton and jute sectors can further contribute to employment generation and foreign exchange earnings, supporting the overall economic development of India.

Question 2

Tea and Coffee plantations should be developed as tourist spots to promote agri-tourism and provide an alternative source of income to the farmers. What do you think about this proposal?

Answer

I believe that developing tea and coffee plantations as tourist spots to promote agri-tourism and provide an alternative source of income to farmers can be a beneficial proposal due to the following reasons:

1. **Tourism Potential** — Tea and coffee plantations offer scenic landscapes, making them attractive to tourists.
2. **Diversification of Income**: Agri-tourism provides an additional source of income for farmers, reducing reliance on agricultural yields alone.
3. **Cultural Exchange and Education** — Visitors can learn about tea and coffee cultivation, processing, and local traditions, enriching their travel experience.
4. **Employment Opportunities** — Developing plantations as tourist spots creates jobs in hospitality, tourism, and service sectors.
5. **Conservation and Sustainability** — Agri-tourism can encourage sustainable practices in plantations, promoting environmental conservation.